

what does test_records need to cover

test_records.py should cover the Task 3 record-layer cases from the assignment. Specifically, it should test: a valid record, modified ciphertext, modified authenticated header, replayed record, and a record reflected into the opposite direction.  HW2-2

A clean breakdown would be:

</> Python

```
def test_valid_record():  
    ...
```

Create a valid record with `seal()`, open it with `open_record()`, and assert the plaintext matches.

</> Python

```
def test_modified_ciphertext_rejected():  
    ...
```

Create a valid record, flip one ciphertext byte, then assert `open_record()` raises an error. Since your HMAC verification uses the cryptography library, this will likely be `InvalidSignature`.

</> Python

```
def test_modified_header_rejected():  
    ...
```

Create a valid record, change something in the authenticated header, such as the message type or sequence byte, then assert it is rejected.

</> Python

```
def test_replay_rejected():  
    ...
```

Open a sequence-0 record once successfully. Then simulate the receiver now expecting sequence 1, but resend the exact same sequence-0 record. It should raise your "Wrong sequence" `ValueError`.